# PROJECT REPORT

## EXPERIMENT - 14

### Project Statement

Design and develop a Worker Health and Safety Monitoring System for Construction Industries. The system monitors worker safety parameters and generates alerts during unsafe working conditions.

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# WORKER HEALTH AND SAFETY MONITORING SYSTEM FOR CONSTRUCTION INDUSTRIES

## Project Overview

The Worker Health and Safety Monitoring System is an IoT-based safety system designed for construction industries. Construction workers may be exposed to excessive temperature, harmful gases, smoke, and health-related risks during their working hours. This system uses sensors connected to an ESP8266 Node MCU to continuously monitor important worker health and environmental parameters.

The system measures the worker's pulse rate using a Pulse Sensor, temperature and humidity using a DHT11 sensor, and gas or smoke concentration using an MQ2 Gas Sensor. A NEO-6M GPS module is used to obtain the worker's location. When an unsafe condition is detected, the ESP8266 activates a buzzer to provide an immediate warning.

## Main Objectives:

The main objectives of the project are:

- To continuously monitor the health condition of construction workers.
- To measure the worker's pulse rate using a Pulse Sensor.
- To monitor temperature and humidity using the DHT11 sensor.
- To detect smoke and harmful/combustible gases using the MQ2 Gas Sensor.
- To identify the worker's real-time location using the NEO-6M GPS module.
- To generate an immediate buzzer alert during unsafe conditions.
- To use the ESP8266 Node MCU for sensor data processing and IoT connectivity.
- To improve worker safety and emergency response at construction sites.

- To reduce dependence on manual safety monitoring.
- To provide a foundation for remote/cloud-based worker safety monitoring.

# Problem Statement

Construction workers are exposed to various health and environmental hazards such as abnormal pulse rate, high temperature, excessive humidity, smoke, and harmful gases. Manual monitoring of these conditions may not provide continuous observation or immediate warning during unsafe situations. In addition, locating a worker during an emergency can be difficult. Therefore, there is a need for an IoT-based Worker Health and Safety Monitoring System that continuously monitors worker health and workplace conditions, detects abnormal situations, provides immediate alerts through a buzzer, and tracks the worker's location using GPS to improve safety and emergency response at construction sites.

## Problems Addressed:

The proposed Worker Health and Safety Monitoring System addresses several important safety challenges in construction industries. It provides continuous monitoring of worker health and surrounding environmental conditions, helping to identify unsafe situations at an early stage.

- **Lack of continuous worker health monitoring:** Monitors the worker's pulse rate continuously.
- **Abnormal health conditions:** Helps identify unusual pulse-rate conditions that may require attention.
- **High temperature and humidity:** Monitors workplace temperature and humidity using the DHT11 sensor.
- **Smoke and harmful gas exposure:** Detects smoke and combustible gases using the MQ2 Gas Sensor.
- **Delayed safety alerts:** Activates the buzzer when predefined unsafe conditions are detected.
- **Difficulty in locating workers:** Uses the NEO-6M GPS module to obtain worker location information.

- **Dependence on manual monitoring:** Automates the collection and monitoring of important safety parameters.
- **Delayed emergency response:** Provides early warnings so that appropriate safety action can be taken quickly.
- **Lack of centralized monitoring:** The ESP8266 provides a foundation for transmitting sensor information to an IoT-based monitoring platform.

# Proposed Solution

The proposed solution is an IoT-based Worker Health and Safety Monitoring System designed to continuously monitor the health of construction workers and their surrounding environmental conditions. The system uses an ESP8266 Node MCU as the main controller to collect and process data from multiple sensors.

A Pulse Sensor monitors the worker's pulse rate, while the DHT11 sensor measures temperature and humidity. An MQ2 Gas Sensor detects smoke and combustible gases in the working environment. The NEO-6M GPS module provides the worker's location information for emergency identification and tracking. When any monitored parameter reaches an unsafe level, the buzzer is activated to provide an immediate warning.

The ESP8266 can also transmit the collected data through Wi-Fi to an IoT platform for remote monitoring and future analysis. This solution helps provide continuous safety monitoring, early hazard detection, immediate alerts, and improved emergency response at construction sites.

## Proposed Operation:

The system operates through the following sequence:

- The ESP8266 Node MCU is powered on and initializes all connected sensors.
- The Pulse Sensor continuously measures the worker's pulse rate.
- The DHT11 Sensor measures the surrounding temperature and humidity.
- The MQ2 Gas Sensor detects smoke and combustible/harmful gases.
- The NEO-6M GPS Module obtains the worker's location information.
- The ESP8266 collects and processes the sensor readings.
- The measured values are compared with predefined safety threshold levels.

- If all parameters are within safe limits, the system continues normal monitoring.
- If an unsafe condition is detected, the buzzer is activated to alert the worker.
- The worker's GPS location can be used to support emergency identification and response.
- Sensor data can be transmitted through Wi-Fi for remote IoT monitoring.
- The system continuously repeats the monitoring process to provide real-time worker safety monitoring.

# System Architecture

The Worker Health and Safety Monitoring System consists of **Input, Processing, Communication, Alert, and Monitoring sections**. The ESP8266 Node MCU acts as the central controller, collecting data from the sensors, analyzing the readings, and generating alerts when unsafe conditions are detected.

## Input Section:

The input section consists of:

- **Pulse Sensor** – Measures the worker's pulse rate.
- **DHT11 Sensor** – Measures temperature and humidity.
- **MQ2 Gas Sensor** – Detects smoke and combustible gases.
- **NEO-6M GPS Module** – Provides worker location information.

These devices collect health, environmental, and location information.

## Processing Section:

- ESP8266 Node MCU acts as the main processing unit.
- Receives data from all connected sensors.
- Processes and compares sensor readings with predefined safety limits.
- Determines whether the worker's condition is safe or unsafe.

## Alert Section:

- **Buzzer** provides an immediate audible warning.

- The buzzer is activated when an unsafe condition is detected, such as high gas concentration or an abnormal monitored value.

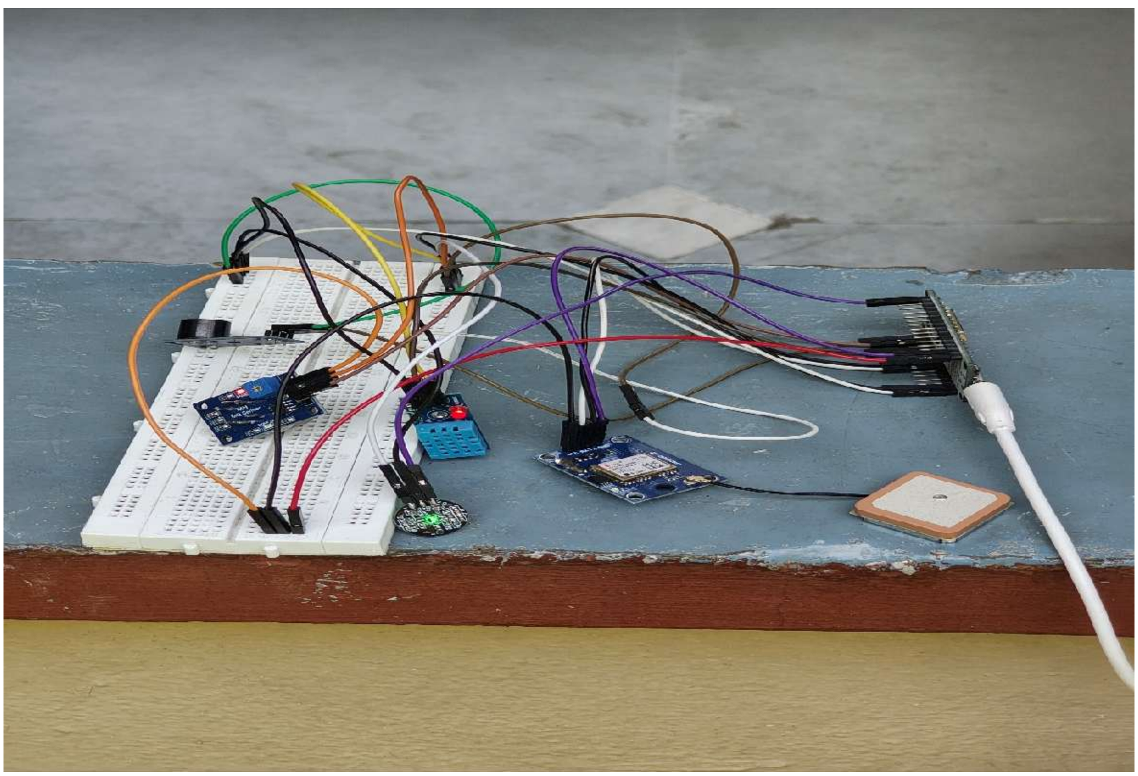
## Location Section:

- The **NEO-6M GPS module** receives satellite signals.
- Provides latitude and longitude information for identifying the worker's location.

## Overall Flow:

Worker → Sensors → ESP8266 → Data Processing → Safety Evaluation → Buzzer / GPS Location

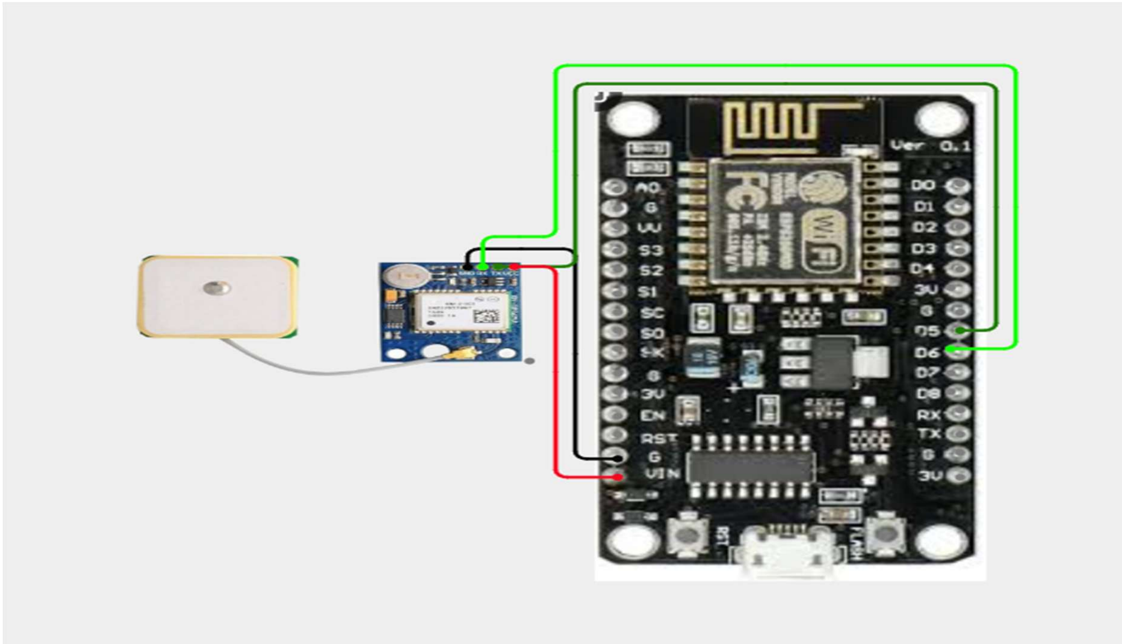
## Image:



## Circuit Table

| S. No. | Component | Component Pin | ESP8266 Node MCU Pin | Function |
|---|---|---|---|---|
| 1 | Pulse Sensor | VCC | 3.3V | Power supply |
| | | GND | GND | Ground |
| | | Signal | A0 / External ADC | Pulse signal |
| 2 | DHT11 Sensor | VCC | 3.3V | Power supply |
| | | GND | GND | Ground |
| | | DATA | D4 (GPIO2) | Temperature & humidity data |
| 3 | MQ2 Gas Sensor | VCC | VIN / Suitable supply | Power supply |
| | | GND | GND | Ground |
| | | DO | D0 (GPIO16) | Gas threshold detection |
| 4 | Buzzer | + | D5 (GPIO14) | Safety alert |
| | | − | GND | Ground |
| 5 | NEO-6M GPS | VCC | Suitable supply | Power supply |
| | | GND | GND | Ground |
| | | TX | D6 (GPIO12) | GPS data to ESP8266 |
| | | RX | D7 (GPIO13) | Data from ESP8266 |
| 6 | Breadboard | Power Rail + | 3.3V / suitable supply | Common power |
| | | Power Rail − | GND | Common ground |

# Circuit Design



## Important Pin Mapping

**Pulse Sensor:**

- VCC → 3.3V
- GND → GND
- Signal → A0 or external ADC input

**DHT11 Sensor:**

- VCC → 3.3V
- GND → GND
- DATA → D4

**MQ2 Gas Sensor:**

- VCC → Suitable supply
- GND → GND
- Digital output → Suitable GPIO for threshold-based detection
- Analog output → External ADC if analog gas measurement is required

**Buzzer:**

- Positive → D5
- Negative → GND

**NEO-6M GPS:**

- VCC → Suitable supply
- GND → GND
- TX → D6
- RX → D7

# Algorithm

- Start the system.
- Initialize the ESP8266 Node MCU and all connected sensors.
- Initialize the Pulse Sensor for worker pulse monitoring.
- Initialize the DHT11 Sensor for temperature and humidity measurement.
- Initialize the MQ2 Gas Sensor for smoke and gas detection.
- Initialize the NEO-6M GPS Module for location tracking.
- Read the pulse rate of the worker.
- Read the temperature and humidity values.
- Read the gas/smoke concentration from the MQ2 sensor.
- Read the GPS location of the worker.
- Process and compare the sensor readings with predefined safety threshold values.
- If all readings are within safe limits, maintain the normal safety status and keep the buzzer OFF.
- If any reading exceeds the specified safety limit, identify the unsafe condition.
- Activate the buzzer to provide an immediate warning.
- Record or transmit the sensor readings and worker location through the ESP8266 Wi-Fi connection.
- Provide the monitored information to the IoT platform or supervisor dashboard, if connected.
- Continue monitoring the worker continuously.
- Repeat the process from Step 7.
- End the process when the system is powered OFF.

# Coding

```
#include <ESP8266WiFi.h>
#include "AdafruitIO_WiFi.h"
#include <DHT.h>
#include <TinyGPS++.h>
#include <SoftwareSerial.h>

// --- Adafruit IO & WiFi Credentials ---
#define WIFI_SSID       "Rithick"
#define WIFI_PASS       "12345678"
#define IO_USERNAME     "Abinaya_1010"
#define IO_KEY          "aio_FIzo20BH06Kz6wRLwFQHWPq3QMTM"

// --- Pin Definitions ---
#define DHTPIN          2   // D4 on NodeMCU
#define DHTTYPE         DHT11
#define MQ2_DIGITAL_PIN 4   // D2 on NodeMCU
#define BUZZER_PIN      5   // D1 on NodeMCU
#define PULSE_PIN       A0

// GPS Software Serial (RX = GPIO 14 / D5, TX = GPIO 12 / D6)
SoftwareSerial gpsSerial(14, 12);

// --- Object Declarations ---
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASS);
DHT dht(DHTPIN, DHTTYPE);
```

```
TinyGPSPlus gps;

// --- Adafruit IO Feeds ---
AdafruitIO_Feed *tempFeed  = io.feed("temperature");
AdafruitIO_Feed *humFeed   = io.feed("humidity");
AdafruitIO_Feed *pulseFeed = io.feed("pulse");
AdafruitIO_Feed *gasFeed   = io.feed("gas-alert");
AdafruitIO_Feed *locFeed   = io.feed("location");

unsigned long lastSend = 0;
const unsigned long sendInterval = 10000; // 10-second interval to avoid rate
throttling

void setup() {
  Serial.begin(115200);
  delay(1000);

  Serial.println("\n============================================
======");
  Serial.println("  WORKER HEALTH & SAFETY MONITORING SYSTEM
(ESP8266)");
  Serial.println("=============================================
=====");

  pinMode(MQ2_DIGITAL_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);
```

```
dht.begin();
gpsSerial.begin(9600);

// --- Wi-Fi Connection Setup ---
Serial.print("[WiFi] Connecting to Network: ");
Serial.println(WIFI_SSID);

WiFi.begin(WIFI_SSID, WIFI_PASS);
while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}

Serial.println("\n[WiFi] Connected Successfully!");
Serial.print("[WiFi] Assigned IP Address: ");
Serial.println(WiFi.localIP());
Serial.print("[WiFi] Signal Strength (RSSI): ");
Serial.print(WiFi.RSSI());
Serial.println(" dBm");

// --- Adafruit IO MQTT Connection ---
Serial.println("\n[Adafruit IO] Connecting to MQTT broker...");
io.connect();

while(io.status() < AIO_CONNECTED) {
```

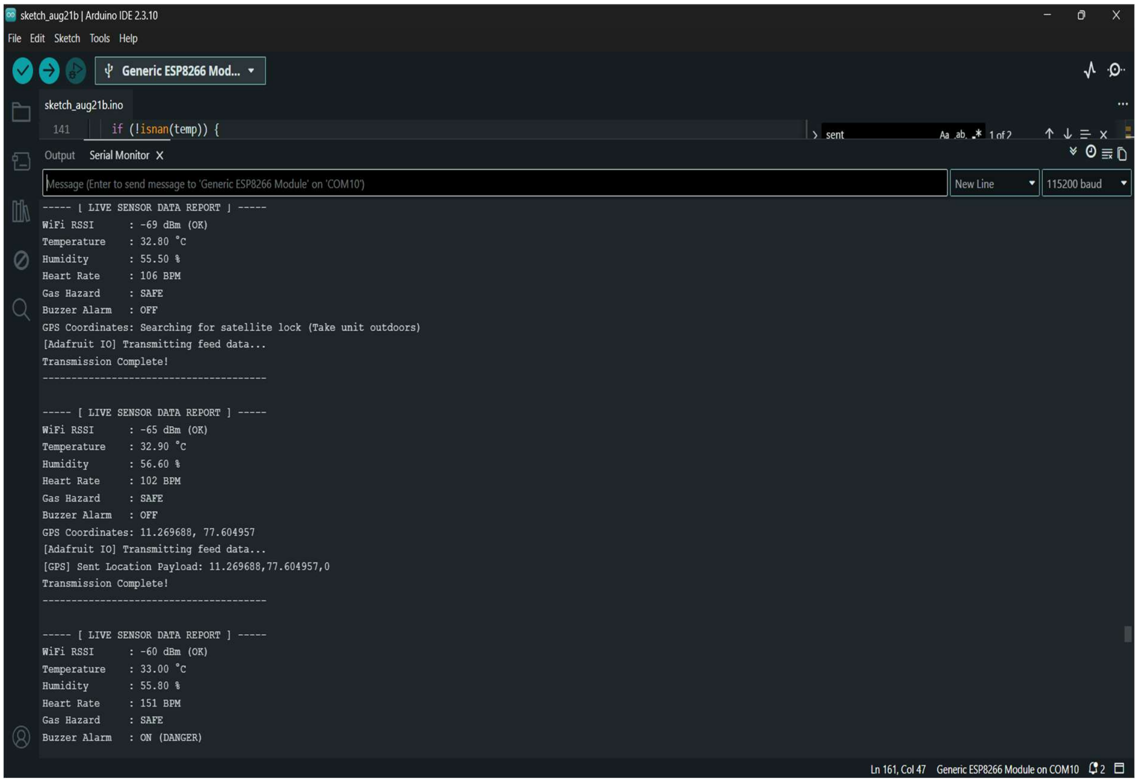
```
    Serial.print(".");
    delay(500);
  }
  Serial.println("\n[Adafruit IO] Connected to Adafruit IO!");
  Serial.println("==========================================
=====\n");
}

void loop() {
  io.run();

  // Continuously parse incoming GPS NMEA sentences
  while (gpsSerial.available() > 0) {
    gps.encode(gpsSerial.read());
  }

  // Scheduled sampling and transmission block
  if (millis() - lastSend > sendInterval) {
    lastSend = millis();

    // 1. Read Sensor Data
    float temp = dht.readTemperature();
    float hum = dht.readHumidity();

    int rawPulse = analogRead(PULSE_PIN);
    int bpm = map(rawPulse, 0, 1023, 0, 180);
```

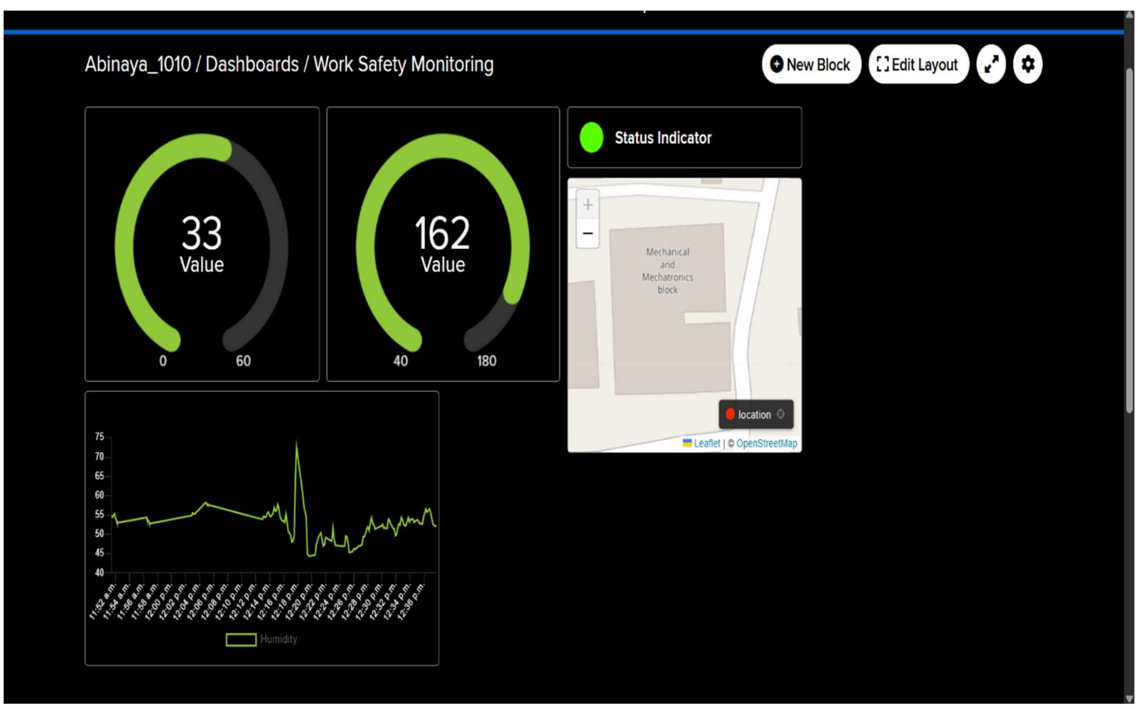
```
  int gasVal = digitalRead(MQ2_DIGITAL_PIN);
  int gasAlert = (gasVal == LOW) ? 1 : 0; // LOW indicates gas threshold
reached

  // 2. Local Safety Alarm Rules
  bool alarmTriggered = false;
  if (gasAlert == 1 || temp > 40.0 || bpm > 120) {
    digitalWrite(BUZZER_PIN, HIGH);
    alarmTriggered = true;
  } else {
    digitalWrite(BUZZER_PIN, LOW);
  }

  // 3. Print Detailed Report to Serial Monitor
  Serial.println("----- [ LIVE SENSOR DATA REPORT ] -----");

  if (WiFi.status() == WL_CONNECTED) {
    Serial.print("WiFi RSSI      : "); Serial.print(WiFi.RSSI()); Serial.println("
dBm (OK)");
  } else {
    Serial.println("WiFi Status    : DISCONNECTED");
  }

  Serial.print("Temperature    : "); Serial.print(temp); Serial.println(" °C");
  Serial.print("Humidity       : "); Serial.print(hum); Serial.println(" %");
  Serial.print("Heart Rate     : "); Serial.print(bpm); Serial.println(" BPM");
```

```
  Serial.print("Gas Hazard     : "); Serial.println(gasAlert == 1 ? "HAZARD DETECTED!" : "SAFE");
  Serial.print("Buzzer Alarm   : "); Serial.println(alarmTriggered ? "ON (DANGER)" : "OFF");

  bool hasGpsLock = gps.location.isValid() && gps.location.age() < 2000;
  if (hasGpsLock) {
    Serial.print("GPS Coordinates: ");
    Serial.print(gps.location.lat(), 6);
    Serial.print(", ");
    Serial.println(gps.location.lng(), 6);
  } else {
    Serial.println("GPS Coordinates: Searching for satellite lock (Take unit outdoors)");
  }

  // 4. Staggered Payload Transmission to Adafruit IO (Avoids Throttle Warnings)
  Serial.println("[Adafruit IO] Transmitting feed data...");

  if (!isnan(temp)) {
    tempFeed->save(temp);
    delay(1200);
  }

  if (!isnan(hum)) {
    humFeed->save(hum);
```

```
    delay(1200);
   }

   pulseFeed->save(bpm);
   delay(1200);

   gasFeed->save(gasAlert);
   delay(1200);

   // Only send location if a valid GPS satellite fix is actively updating
   if (hasGpsLock) {
    char locBuffer[64];
    // Adafruit IO Location CSV Format: value,latitude,longitude,elevation
    snprintf(locBuffer, sizeof(locBuffer), "%.6f,%.6f,0",
             gps.location.lat(), gps.location.lng());
    locFeed->save(locBuffer);
    Serial.print("[GPS] Sent Location Payload: ");
    Serial.println(locBuffer);
   }

   Serial.println("Transmission Complete!");
   Serial.println("---------------------------------------\n");
  }
}
```

# Coding Output:



## System Working

The Worker Health and Safety Monitoring System continuously monitors the health condition of a construction worker and the surrounding environmental conditions. The ESP8266 Node MCU acts as the main controller and collects data from all connected sensors.

- The Pulse Sensor continuously measures the worker's pulse rate and provides the reading to the ESP8266.
- The DHT11 Sensor measures the surrounding temperature and humidity.
- The MQ2 Gas Sensor detects smoke and combustible gases in the workplace.
- The NEO-6M GPS Module receives satellite signals and provides the worker's location information.
- The ESP8266 Node MCU collects and processes the sensor readings.
- The controller compares the measured values with predefined safety threshold levels.
- When the readings are within the safe range, the system maintains normal monitoring and the buzzer remains OFF.
- If an unsafe condition is detected, such as an abnormal monitored value or high gas/smoke level, the buzzer is activated to alert the worker.
- The worker's GPS location can be used to support emergency identification and response.
- Through Wi-Fi, the ESP8266 can transmit sensor information to an IoT/cloud platform for remote monitoring.
- The system continuously repeats the monitoring process, allowing unsafe
- conditions to be detected as early as possible.

# Bill of Materials

| S. No. | Component | Specification / Model | Quantity | Purpose |
|---|---|---|---|---|
| 1 | ESP8266 Node MCU | Node MCU ESP8266 Wi-Fi Development Board | 1 | Main controller and data processing |
| 2 | Pulse Sensor | Pulse/Heart Rate Sensor | 1 | Measures worker's pulse rate |
| 3 | DHT11 Sensor | Digital Temperature & Humidity Sensor | 1 | Measures temperature and humidity |
| 4 | MQ2 Gas Sensor | MQ2 Gas/Smoke Detection Module | 1 | Detects smoke and combustible gases |
| 5 | Buzzer | 5V Active Buzzer | 1 | Provides audible safety alert |
| 6 | GPS Module | NEO-6M GPS Module | 1 | Determines worker's geographical location |
| 7 | Breadboard | Standard 400/830-point Breadboard | 1 | Circuit assembly |
| 8 | Jumper Wires | Male-Male / Male-Female | 1 Set | Connecting components |
| 9 | USB Cable | Micro-USB | 1 | Programming and powering Node MCU |
| 10 | External ADC / Analog Interface | Optional, e.g. ADS1115 | 1 | Allows additional analog sensor measurement if required |
| 11 | Power Supply | 5V USB Power Source / Power Bank | 1 | Portable system power |

# Prototype Implementation

The prototype of the Worker Health and Safety Monitoring System is implemented using an ESP8266 Node MCU, Pulse Sensor, DHT11 Sensor, MQ2 Gas Sensor, Buzzer, and NEO-6M GPS Module. All components are assembled on a breadboard and interconnected using jumper wires.

The ESP8266 Node MCU is used as the main controller. The Pulse Sensor is connected to monitor the worker's pulse rate, while the DHT11 Sensor measures temperature and humidity. The MQ2 Gas Sensor is connected to detect smoke and combustible gases in the surrounding environment. The NEO-6M GPS Module is connected to obtain the worker's location, and the Buzzer is connected to provide an audible warning during unsafe conditions.

The control program is developed and uploaded to the ESP8266 using the Arduino IDE. The program continuously reads the sensor values and compares them with predefined safety limits. When all parameters are within the safe range, the system continues normal monitoring. If an unsafe condition is detected, the ESP8266 activates the buzzer and identifies the corresponding safety condition. GPS information can also be used for worker location monitoring.

## Hardware Setup:

The hardware setup of the Worker Health and Safety Monitoring System consists of the ESP8266 Node MCU, Pulse Sensor, DHT11 Sensor, MQ2 Gas Sensor, Buzzer, NEO-6M GPS Module, breadboard, and jumper wires.

- **ESP8266 Node MCU:** Acts as the main controller and processes the data received from the sensors.
- **Pulse Sensor:** Connected to the worker's finger to measure pulse-rate signals continuously.
- **DHT11 Sensor:** Measures the surrounding temperature and humidity at the construction site.
- **MQ2 Gas Sensor:** Detects smoke and combustible gases present in the surrounding environment.
- **NEO-6M GPS Module:** Receives GPS signals and provides the worker's geographical location.

- **Buzzer:** Connected to a digital output of the ESP8266 to provide an audible warning when an unsafe condition is detected.
- **Breadboard:** Used to mount and interconnect the electronic components during prototype development.
- **Jumper Wires:** Used to establish power, ground, and signal connections between the ESP8266 and sensors.

# Prototype Working

The prototype starts when the ESP8266 Node MCU is powered on. The controller initializes the Pulse Sensor, DHT11 Sensor, MQ2 Gas Sensor, NEO-6M GPS Module, and Buzzer. After initialization, the system continuously collects health, environmental, and location data from the connected sensors.

The Pulse Sensor monitors the worker's pulse rate, while the DHT11 measures temperature and humidity. The MQ2 Gas Sensor checks for smoke and combustible gases in the surrounding environment. At the same time, the NEO-6M GPS Module obtains the worker's location information.

The ESP8266 processes these readings and compares them with the predefined safety limits. Under normal conditions, the system maintains a safe status and the buzzer remains OFF. If an unsafe condition is detected, such as an abnormal monitored value or high gas/smoke level, the ESP8266 activates the buzzer to provide an immediate warning.

The GPS information can be used to identify the worker's location during an emergency, while the ESP8266's Wi-Fi capability can be used to transmit monitoring data to an IoT platform. The system continuously repeats this process, enabling real-time monitoring of worker health and workplace safety conditions.

# Expected Prototype Output

The prototype is expected to continuously monitor the worker's health, environmental conditions, and location. The ESP8266 processes the sensor readings and provides an appropriate safety status.

# Prototype Demonstration

The prototype demonstration is carried out to verify the operation of the Worker Health and Safety Monitoring System under normal and unsafe conditions.

- **Power ON the System:** Power the ESP8266 Node MCU and verify that all connected sensors are initialized successfully.
- **Pulse Monitoring Demonstration:** Place the Pulse Sensor on the worker's finger and observe the pulse-rate readings received by the ESP8266.
- **Temperature and Humidity Demonstration:** Operate the DHT11 sensor under normal environmental conditions and observe the temperature and humidity values.
- **Gas Detection Demonstration:** The MQ2 Gas Sensor is used in a controlled test environment to demonstrate smoke/gas detection. When the reading increases beyond the configured threshold, the system identifies a warning condition.
- **GPS Demonstration:** Place the NEO-6M GPS module where it has a suitable view of the sky and demonstrate the acquisition of the worker's location when a valid satellite fix is obtained.
- **Safety Alert Demonstration:** When a monitored parameter crosses its configured safety limit, the ESP8266 activates the **buzzer** to provide an immediate audible warning.
- **Normal Condition Demonstration:** When all monitored values remain within the configured safe ranges, the system displays/maintains a **SAFE** status and keeps the buzzer OFF.

- **IoT Monitoring Demonstration:** If an IoT platform is connected, demonstrate the transmission of sensor readings through the ESP8266 Wi-Fi connection for remote monitoring.
- **Emergency Response Demonstration:** Demonstrate how the warning condition and available GPS location information can assist a supervisor in identifying the worker's situation and responding appropriately

## Results & Performance Analysis

The developed Worker Health and Safety Monitoring System successfully demonstrates continuous monitoring of important worker health and workplace safety parameters. The ESP8266 Node MCU collects and processes data from the Pulse Sensor, DHT11 Sensor, MQ2 Gas Sensor, and NEO-6M GPS Module.

The system successfully monitors the worker's pulse rate, temperature, humidity, and gas/smoke conditions. The GPS module provides location information when a valid satellite fix is available. When a monitored parameter exceeds its configured safety threshold, the ESP8266 activates the buzzer to provide an immediate warning.

The prototype demonstrates that an IoT-based system can provide continuous safety monitoring and support faster identification of potentially unsafe working conditions.

## Performance Analysis

| Parameter | Observation | Performance |
|---|---|---|
| Pulse Monitoring | Pulse signal is continuously monitored | Good |
| Temperature Monitoring | DHT11 provides temperature readings | Good |
| Humidity Monitoring | Humidity is monitored continuously | Good |
| Gas/Smoke Detection | MQ2 detects increased smoke/gas levels | Good |
| GPS Tracking | Provides location after obtaining a valid GPS fix | Good |
| Alert System | Buzzer activates during configured unsafe conditions | Good |
| Data Processing | ESP8266 processes sensor readings effectively | Good |
| Wi-Fi Connectivity | Supports IoT-based remote monitoring | Good |
| Continuous Monitoring | System repeatedly checks sensor conditions | Good |